usepackage[a4paper,margin=0.65in]geometry

CAYLEY, Collected Mathematical Papers, Vol. II.

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Papers 108–114.

108.

ON CERTAIN MULTIPLE INTEGRALS CONNECTED WITH THE THEORY OF ATTRACTIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174–178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," *Journal*, t. ii. [1847], pp. 219–223, [44], the equation

$$\int \frac{d\xi\, d\eta \cdots}{\left[(\xi-\alpha)^2+(\eta-\beta)^2+\cdots+v^2\right]^{\frac{1}{2}n-q}} \;=\; {\theta_1}^q\,\Gamma\left(\tfrac{1}{2}n-q\right)\Gamma(q+1)\int_e^\infty \frac{s^{n-q-1}\,({\theta_1}^2-\sigma)^q\,ds}{\sqrt{\left(s+\frac{f^2}{{\theta_1}^2}\right)\left(s+\frac{g^2}{{\theta_1}^2}\right)\cdots}}\,,$$

where n is the number of variables of the multiple integral, and the condition of the integration is

$$\frac{(\xi-\alpha)^2}{f^2}+\frac{(\eta-\beta)^2}{g^2}+\cdots \;\leqq\; 1\,;$$

also where

$$\sigma \;=\; \frac{(\alpha-\alpha_1)^2}{s+\frac{f^2}{{\theta_1}^2}} \;+\; \frac{(\beta-\beta_1)^2}{s+\frac{g^2}{{\theta_1}^2}} \;+\;\cdots\;+\; \frac{v^2}{s}\,,$$

and e is the positive root of

$${\theta_1}^2 \;=\; \frac{(\alpha-\alpha_1)^2}{e+\frac{f^2}{{\theta_1}^2}} \;+\; \frac{(\beta-\beta_1)^2}{e+\frac{g^2}{{\theta_1}^2}} \;+\;\cdots\;+\; \frac{v^2}{e}\,.$$

Suppose $f=g=\cdots=\theta_1$, and write $(\alpha-\alpha_1)^2+\cdots=k^2$, we obtain

$$\int \frac{d\xi\, d\eta \cdots}{\left[(\xi-\alpha)^2+\cdots+v^2\right]^{\frac{1}{2}n-q}} \;=\; \frac{\pi^{\frac{1}{2}n}}{\Gamma\left(\frac{1}{2}n-q\right)\Gamma(q+1)}\int_e^\infty \frac{s^{q-1}\,({\theta_1}^2-\sigma)^q\,ds}{(1+s)^{\frac{1}{2}n}}\,,$$

the limiting condition for the multiple integral being

$$(\xi-\alpha_1)^2+\cdots \;\leqq\; {\theta_1}^2,$$

and the function σ, and limit e, being given by

$$\sigma \;=\; \frac{(\alpha-\alpha_1)^2}{1+s}+\frac{(\beta-\beta_1)^2}{1+s}+\cdots+\frac{v^2}{s}\,, \qquad {\theta_1}^2 \;=\; \frac{(\alpha-\alpha_1)^2}{1+e}+\frac{(\beta-\beta_1)^2}{1+e}+\cdots+\frac{v^2}{e}\,,$$

e denoting, as before, the positive root. Observing that the quantity under the integral sign on the second side vanishes for $s=e$, there is no difficulty in deducing, by a differentiation with respect to θ_1, the formula

$$\int \frac{d\Sigma}{\left[(\xi-\alpha)^2+\cdots+v^2\right]^{\frac{1}{2}n-q+1}} \;=\; \frac{2{\theta_1}^{q+1}\pi^{\frac{1}{2}n}}{\Gamma\left(\frac{1}{2}n-q\right)\Gamma(q)}\int_e^\infty \frac{s^{q-1}\,({\theta_1}^2-\sigma)^{q-1}\,ds}{(1+s)^{\frac{1}{2}n}}\,,$$

in which e is the positive root of the equation

$$\theta_1{}^2 s^2 + \chi s - v^2 \;=\; 0.$$

I propose to transform these formulæ by means of the theory of images; it will be convenient to investigate some preliminary formulæ. Suppose $\lambda^2 = \alpha^2 + \beta^2 + \cdots$, $\lambda_1{}^2 = \alpha_1{}^2 + \beta_1{}^2 + \cdots$; also consider the new constants $a, b, \ldots,\ a_1, b_1, \ldots,\ u, f_1$, determined by the equations

$$\frac{\partial^2 u}{\partial\lambda^2 + v^2} = \alpha, \qquad \frac{\partial^2 u_1}{\lambda_1{}^2 - f_1{}^2} = a_1, \qquad \ldots \qquad \frac{\partial^2 v}{\lambda^2 + v^2} = u, \quad \frac{\partial^2 f_1}{\lambda_1{}^2 - f_1{}^2} = f_1\,,$$

where S is arbitrary. Then, putting

$$l^2 = a^2 + b^2 + \cdots\,, \qquad l_1{}^2 = a_1{}^2 + b_1{}^2 + \cdots\,,$$

it is easy to see that

$$(\lambda^2 + v^2)(l^2 + u^2) \;=\; S^2, \qquad (\lambda_1{}^2 - f_1{}^2)(l_1{}^2 - f_1{}^2) \;=\; S^2\alpha_1{}^2/\alpha^2,$$

and

$$\frac{\partial^2 u}{l^2 + u^2} \;=\; \lambda, \qquad \frac{\partial^2 f_1}{l_1{}^2 - f_1{}^2} \;=\; \lambda_1\,,$$

and so on.

Proceeding to express the single integrals in terms of the new constants, we have in the first place $k^2 = \theta^2 k'^2$, where

$$k'^2 \;=\; \frac{p^2}{(l^2 + u^2)} + \frac{l_1{}^2}{(l_1{}^2 - f_1{}^2)^2} + \cdots\,;$$

or if we write

$$aa_1 + bb_1 + \cdots \;=\; l\,l_1\,\cos\omega,$$

we have

$$k'^2 \;=\; \frac{l^2}{l^2 + u^2} + \frac{l_1{}^2}{l_1{}^2 - f_1{}^2} - \frac{2\,l\,l_1 \cos\omega}{(l^2 + u^2)^{\frac{1}{2}}(l_1{}^2 - f_1{}^2)^{\frac{1}{2}}}\,.$$

Hence also $\chi = \theta^2\chi'$, where

$$\chi' \;=\; \frac{f_1{}^2}{(l_1{}^2 - f_1{}^2)^2} - k'^2 - \frac{u^2}{(l^2 + u^2)^2}\,,$$

whence

$$-\chi' \;=\; \frac{1}{l^2+u^2}+\frac{1}{{l_1}^2-{f_1}^2}+\frac{2\,l\,l_1\cos\omega}{(l^2+u^2)^{\frac12}({l_1}^2-{f_1}^2)^{\frac12}} \;=\; \frac{1}{(l^2+u^2)({l_1}^2-{f_1}^2)}\big[\,p^2+u^2-{f_1}^2\,\big],$$

where $p^2 = l^2+{l_1}^2-2\,l\,l_1\cos\omega$, that is

$$p^2 \;=\; (a-a_1)^2+(b-b_1)^2+\cdots;$$

consequently ${\theta_1}^2s^2+\chi s-v^2 = -\theta^2S^2\,\Pi$, where Π is given by

$$\Pi \;=\; \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2}\,s^2-\frac{(p^2+u^2-{f_1}^2)}{(l^2+u^2)({l_1}^2-{f_1}^2)}\,s-\frac{u^2}{(l^2+u^2)^2}\,,$$

and it is clear that e will be the positive root of

$$0 \;=\; \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2}\,e^2-\frac{(p^2+u^2-{f_1}^2)}{(l^2+u^2)({l_1}^2-{f_1}^2)}\,e-\frac{u^2}{(l^2+u^2)^2}\,.$$

It may be noticed that, in the particular case of $u=0$, the roots of this equation are 0 and $\dfrac{(p^2-{f_1}^2)({l_1}^2-{f_1}^2)}{{l_1}^2{f_1}^2}$. Consequently if $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of opposite signs, we have $e=0$; but if $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of the same sign, $e=\dfrac{(p^2-{f_1}^2)({l_1}^2-{f_1}^2)}{{l_1}^2{f_1}^2}$.

In order to transform the double integrals, considering the new variables $x, y, \ldots$, I write $x^2+y^2+\cdots = r^2$ and

$$\xi \;=\; \frac{Sx}{r^2}, \quad \ldots$$

whence also, if $\xi^2+\eta^2+\cdots=\rho^2$ (which gives $r\rho = S$), we have

$$\alpha \;=\; \frac{d\xi}{d\rho^2}\,, \quad \ldots\,;$$

also it is immediately seen that

$$(\xi-\alpha)^2+\cdots+v^2 \;=\; \frac{S^2}{(l^2+u^2)\,r^2}\big[(x-a)^2+\cdots+u^2\big],$$

$$(\xi-\alpha_1)^2+\cdots-{\theta_1}^2 \;=\; \frac{S^2}{({l_1}^2-{f_1}^2)\,r^2}\big[(x-a_1)^2+\cdots-{f_1}^2\big];$$

and from the latter equation it follows that the limiting condition for the first integral is $(x-a_1)^2+\cdots \leqq {f_1}^2$ (there is no difficulty in seeing that the sign $<$ in the former limiting condition gives rise here to the sign $>$), and that the second integral has to be extended over the surface $(x-a_1)^2+\cdots = {f_1}^2$. Also if dS represent the element of this surface, we may obtain

$$d\xi\, d\eta\, \cdots \;=\; \frac{S^n}{r^{2n}}\, dx\, dy\, \cdots\,, \qquad d\Sigma \;=\; \frac{S^{n-1}}{r^{2n-2}}\, dS\,;$$

and, combining the above formulæ, we obtain

$$\int \frac{dx\, dy\, \cdots}{(x^2+y^2+\cdots)^{\frac{1}{2}n+q}\left[(x-a)^2+(y-b)^2+\cdots+u^2\right]^{\frac{1}{2}n-q}} \;=\; \frac{\pi^{\frac{1}{2}n}}{\Gamma\big(\tfrac{1}{2}n-q\big)\,\Gamma(q+1)\,(l^2+u^2)^{\frac{1}{2}n-q}} \int_e^{\infty} \frac{}{s}$$

the limiting condition of the multiple integral being

$$(x-a_1)^2+(y-b_1)^2+\cdots \;\leqq\; {f_1}^2\,;$$

and

$$\int \frac{dS}{(x^2+y^2+\cdots)^{\frac{1}{2}n+q-1}\left[(x-a)^2+(y-b)^2+\cdots+u^2\right]^{\frac{1}{2}n-q}} \;=\; \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma\big(\tfrac{1}{2}n-q\big)\,\Gamma(q)\,(l^2+u^2)^{\frac{1}{2}n-q}\,({l_1}^2-f}$$

where dS is the element of the surface $(x-a_1)^2+(y-b_1)^2+\cdots = {f_1}^2$, and the integration extends over the entire surface. In these formulæ, l, l_1, p, Π denote as follows:

$$l^2 = a^2+b^2+\cdots\,, \quad {l_1}^2 = {a_1}^2+{b_1}^2+\cdots\,, \quad p^2 = (a-a_1)^2+(b-b_1)^2+\cdots\,,$$

$$\Pi \;=\; \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2}\, s^2 - \frac{(p^2+u^2-{f_1}^2)}{(l^2+u^2)\,({l_1}^2-{f_1}^2)}\, s - \frac{u^2}{(l^2+u^2)^2}\,,$$

and e is the positive root of the equation $\Pi = 0$.

The only obviously integrable case is that for which in the second formula $q=1$; this gives

$$\int \frac{dS}{(x^2+y^2+\cdots)^{\frac{1}{2}n}\left[(x-a)^2+(y-b)^2+\cdots+u^2\right]^{\frac{1}{2}n-1}} \;=\; \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma\big(\tfrac{1}{2}n\big)\,(l^2+u^2)^{\frac{1}{2}n-1}\,({l_1}^2-{f_1}^2)\,(1+e)^{\frac{1}{2}n-}}$$

In the case of $u=0$, we have, as before, when $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of opposite signs, $e=0$, and therefore $1+e=1$; but when $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of the same sign, the value before found for e gives

$$1+e \;=\; \frac{1}{{l_1}^2{f_1}^2}\left[\,{l_1}^2{f_1}^2+(p^2-{f_1}^2)({l_1}^2-{f_1}^2)\,\right].$$

Consider the image of the origin with respect to the sphere $(x-a_1)^2+(y-b_1)^2+\cdots = {f_1}^2$; the coordinates of this image are

$$\frac{a_1}{{l_1}^2}({l_1}^2-{f_1}^2), \qquad \frac{b_1}{{l_1}^2}({l_1}^2-{f_1}^2), \quad \ldots,$$

and consequently, if μ be the distance of this image from the point $(a, b, \ldots)$, we have

$$\mu^2 \;=\; \bigl[a - \tfrac{a_1}{{l_1}^2}({l_1}^2-{f_1}^2)\bigr]^2 + \cdots \;=\; \frac{1}{{l_1}^2}\bigl[\,l^2{f_1}^2 + (p^2-{f_1}^2)({l_1}^2-{f_1}^2)\,\bigr];$$

whence, by a simple reduction,

$$1+e \;=\; \frac{{l_1}^2\mu^2}{{l_1}^2{f_1}^2}\,,$$

or the values of the integral are

$$p^2-{f_1}^2 \text{ and } {l_1}^2-{f_1}^2 \text{ opposite signs,} \quad I \;=\; \frac{2\pi^{\frac{1}{2}n}}{\Gamma\bigl(\frac{1}{2}n\bigr)}\cdot\frac{f_1}{l^{n-2}({l_1}^2-{f_1}^2)}\,,$$

$$p^2-{f_1}^2 \text{ and } {l_1}^2-{f_1}^2 \text{ the same sign,} \quad I \;=\; \frac{2\pi^{\frac{1}{2}n}}{\Gamma\bigl(\frac{1}{2}n\bigr)}\cdot\frac{{f_1}^{n-1}}{l^{n-2}\,\mu^{n-2}({l_1}^2-{f_1}^2)}\,,$$

where μ is the distance from the point $(a, b, \ldots)$ of the image of the origin with respect to the sphere $(x-a_1)^2+\cdots-{f_1}^2=0$.

Stone Buildings, August 6, 1850.

109.

ON THE RATIONALISATION OF CERTAIN ALGEBRAICAL EQUATIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. viii. (1853), pp. 97–101.]

Suppose

$$x + y = 0, \qquad x^2 = a, \qquad y^2 = b\,;$$

then if we multiply the first equation by 1, xy, and reduce by the two others, we have

$$x + y = 0, \qquad bx + ay = 0,$$

from which, eliminating x, y,

$$\begin{vmatrix} 1 & 1 \\ b & a \end{vmatrix} = 0\,;$$

which is the equation between a and b; or, considering x, y as quadratic radicals, the rational equation between x, y. So if the original equation be multiplied by x, y, we have

$$a + xy = 0, \qquad b + xy = 0\,;$$

or, eliminating 1, xy,

$$\begin{vmatrix} a & 1 \\ b & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between x, y.

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1, yz, zx, xy, reduce and eliminate the quantities x, y, z, xyz, we have the rational equation

$$\begin{vmatrix} 1 & 1 & 1 & \cdot \\ \cdot & c & b & \cdot \\ c & \cdot & a & \cdot \\ b & a & \cdot & \cdot \end{vmatrix} = 0\,;$$

and again, multiply the equation by x, y, z, xyz, reduce and eliminate the quantities 1, yz, zx, xy, the result is

$$\begin{vmatrix} a & b & c & \cdot \\ a & \cdot & 1 & 1 \\ b & 1 & \cdot & 1 \\ c & 1 & 1 & \cdot \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper "On a theorem in the Geometry of Position" (*Journal*, vol. ii. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by 1, yz, zx, xy, xw, yw, zw, $xyzw$, reducing and eliminating the quantities in the outside row,

$$x,\ y,\ z,\ w,\ yzw,\ zwx,\ wxy,\ xyz,$$

we have the result

$$\begin{vmatrix} 1 & 1 & 1 & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & c & b & \cdot & \cdot & 1 & \cdot & 1 \\ c & \cdot & a & \cdot & 1 & \cdot & \cdot & 1 \\ b & a & \cdot & \cdot & \cdot & \cdot & 1 & 1 \\ d & \cdot & \cdot & a & \cdot & 1 & 1 & \cdot \\ \cdot & d & \cdot & b & 1 & \cdot & 1 & \cdot \\ \cdot & \cdot & d & c & 1 & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by x, y, z, w, yzw, zwx, wxy, and xyz, reduce and eliminate the quantities in the outside row,

$$1,\ yz,\ zx,\ xy,\ xw,\ yw,\ zw,\ xyzw,$$

we have the result

$$\begin{vmatrix} a & \cdot & \cdot & \cdot & 1 & 1 & 1 & \cdot \\ b & \cdot & 1 & 1 & \cdot & 1 & \cdot & \cdot \\ c & 1 & \cdot & 1 & 1 & \cdot & \cdot & \cdot \\ d & \cdot & \cdot & \cdot & 1 & 1 & 1 & \cdot \\ \cdot & d & \cdot & \cdot & \cdot & c & b & 1 \\ \cdot & \cdot & d & \cdot & c & \cdot & a & 1 \\ \cdot & \cdot & \cdot & d & b & a & \cdot & 1 \\ \cdot & a & b & c & \cdot & \cdot & \cdot & 1 \end{vmatrix} = 0,$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns.

And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x + y = 0, \qquad x^3 = a, \qquad y^3 = b\,;$$

and multiply first by 1, x^2y, xy^2; this gives

$$x + y = 0, \qquad x^2y + xy^2 = 0, \qquad ay + x^2y^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & \cdot \\ \cdot & \cdot & 1 \\ \cdot & a & 1 \\ b & \cdot & 1 \end{vmatrix} = 0\,;$$

[*the original prints a* 3×3 *array of dots and units corresponding to the elimination scheme; the determinant condition is what is significant.*]

next multiply by x, y, x^2y^2; this gives

$$x^2 + xy = 0, \qquad y^2 + xy = 0, \qquad bx^2 + ay^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} 1 & \cdot & 1 \\ \cdot & 1 & 1 \\ b & a & \cdot \end{vmatrix} = 0\,;$$

and lastly, multiply by x^2, y^2, xy; this gives

$$a + x^2y = 0, \qquad b + xy^2 = 0, \qquad x^2y + xy^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} a & 1 & \cdot \\ b & \cdot & 1 \\ \cdot & 1 & 1 \end{vmatrix} = 0,$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.

Applying the preceding process to the system

$$x + y + z = 0, \qquad x^3 = a, \qquad y^3 = b, \qquad z^3 = c,$$

multiply first by 1, xyz, $x^2y^2z^2$, x^2z, y^2x, z^2y, x^2y, y^2z, z^2x, reduce and eliminate the quantities in the outside row,

$$x,\ y,\ z,\ y^2z^2,\ z^2x^2,\ x^2y^2,\ x^2yz,\ y^2zx,\ z^2xy,\ z^2x^2,\ x^2y^2,$$

the result is a determinantal equation

$$\Delta_1 = 0$$

[whose schematic form, written out by Cayley as a 9×9 array of 1's, $\cdot$'s, and a, b, c's, is — for legibility — abridged here; see facsimile, p. 43].

next multiply by x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy, reduce and eliminate the quantities in the outside row,

$$x^2,\ y^2,\ z^2,\ yz,\ zx,\ xy,\ xy^2z^2,\ yz^2x^2,\ zx^2y^2,$$

the result is a second determinantal equation $\Delta_2 = 0$ of the same size;

lastly, multiply by x^2, y^2, z^2, yz, zx, xy, $x^2y^2z^2$, $y^2z^2x^2$, $x^2y^2z^2$, reduce and eliminate the quantities in the outside row,

$$1,\ x^2y^2,\ y^2z^2,\ z^2x^2,\ x^2yz,\ y^2zx,\ z^2xy,\ xy^2z,\ yz^2x,$$

the result is a third determinantal equation $\Delta_3 = 0$:

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

[The three large determinantal arrays printed at pp. 43–44 of the original are partially illegible in the present scan and have been replaced by their schematic descriptions; the substantive content — the existence and equivalence relations — is preserved.]

110.

NOTE ON THE TRANSFORMATION OF A TRIGONOMETRICAL EXPRESSION.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 61–62.]

THE differential equation

$$\frac{dx}{(a+x)\sqrt{c+x}} + \frac{dy}{(a+y)\sqrt{c+y}} + \frac{dz}{(a+z)\sqrt{c+z}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1}\sqrt{\frac{a-c}{c+x}} + \tan^{-1}\sqrt{\frac{a-c}{c+y}} + \tan^{-1}\sqrt{\frac{a-c}{c+z}} = 0\,;$$

and, by Abel's theorem,

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = 0.$$

To show *à posteriori* the equivalence of these two equations, I represent the determinant by the symbol $\square$, and expressing it in the form

$$\square = \begin{vmatrix} 1, & a+x, & (a+x)\sqrt{c+x} \\ 1, & a+y, & (a+y)\sqrt{c+y} \\ 1, & a+z, & (a+z)\sqrt{c+z} \end{vmatrix},$$

I write for the moment $\xi = \sqrt{\dfrac{a-c}{c+x}}$, $\eta = \sqrt{\dfrac{a-c}{c+y}}$, $\zeta = \sqrt{\dfrac{a-c}{c+z}}$, so that

$$(a+x) = (a-c)\Big(1+\tfrac{1}{\xi^2}\Big), \qquad (a+x)\sqrt{c+x} = (a-c)^{\frac{3}{2}}\Big(\tfrac{1}{\xi^3}+\tfrac{1}{\xi}\Big),$$

and similarly for $(a+y)$, $(a+z)$, etc. Then

$$\square = \frac{(a-c)^{\frac{5}{2}}}{\xi^3\eta^3\zeta^3}\begin{vmatrix} \xi^2, & \xi^2+1, & \xi^2+1 \\ \eta^2, & \eta^2+1, & \eta^2+1 \\ \zeta^2, & \zeta^2+1, & \zeta^2+1 \end{vmatrix}$$

$$= \frac{(a-c)^{\frac{5}{2}}}{\xi^3\eta^3\zeta^3}\,(\xi+\eta+\zeta-\xi\eta\zeta)\begin{vmatrix} 1, & \xi, & \xi^2 \\ 1, & \eta, & \eta^2 \\ 1, & \zeta, & \zeta^2 \end{vmatrix};$$

or, replacing ξ, η, ζ by their values, we have identically

$$\begin{vmatrix} 1, & x, & (a+x)\sqrt{c+x} \\ 1, & y, & (a+y)\sqrt{c+y} \\ 1, & z, & (a+z)\sqrt{c+z} \end{vmatrix} = K(x,y,z;a,c)\Big[\sqrt{\tfrac{a-c}{c+x}}+\sqrt{\tfrac{a-c}{c+y}}+\sqrt{\tfrac{a-c}{c+z}}-\sqrt{\tfrac{a-c}{c+x}}\sqrt{\tfrac{a-c}{c+y}}\sqrt{\tfrac{a-c}{c+z}}\Big],$$

where $K(x,y,z;a,c)$ is the symmetric prefactor displayed above; and the equation

$$\sqrt{\tfrac{a-c}{c+x}}+\sqrt{\tfrac{a-c}{c+y}}+\sqrt{\tfrac{a-c}{c+z}}-\sqrt{\tfrac{a-c}{c+x}}\sqrt{\tfrac{a-c}{c+y}}\sqrt{\tfrac{a-c}{c+z}} = 0$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1}\sqrt{\tfrac{a-c}{c+x}}+\tan^{-1}\sqrt{\tfrac{a-c}{c+y}}+\tan^{-1}\sqrt{\tfrac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

111.

ON A THEOREM OF M. LEJEUNE-DIRICHLET'S.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 163–165.]

THE following formula,

$$\Sigma\, q^{ax^2+2bxy+cy^2} . \Sigma\, q^{a'x'^2+2b'x'y'+c'y'^2} \cdots \;=\; 2\,\Sigma \left[S^{n-c}\, \epsilon^{\frac{1}{2}(n-1)}\right] \left(\tfrac{n}{P}\right)\, q^{nn'\cdots}, \qquad (3)$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. xxi. [1840] p. 8). The notation is as follows:—On the left-hand side (a, b, c), (a', b', c'), $\ldots$ are a system of properly primitive forms to the negative determinant $-D$ (i.e. a system of positive forms); x, y are positive or negative integers including zero, such that in the sum $\Sigma\, q^{ax^2+2bxy+cy^2}$, $ax^2 + 2bxy + cy^2$ is prime to $2D$, and similarly in the other sums; q is indeterminate and the summations extend to the values first mentioned, of x and y. On the right-hand side we have to consider the form of D, viz. we have $D = PS^2$ or else $D = 2PS^2$, where S^2 is the greatest square factor in D and where P is odd: this obviously defines P, and the values of S, ϵ, which are always ± 1 (or, as I prefer to express it, are always $\pm$) are given as follows, viz.

$$\begin{array}{lll}
D = PS^2, & P \equiv 1 \pmod 4, & S, \epsilon = +, +, \\
D = PS^2, & P \equiv 3 \pmod 4, & S, \epsilon = -, +, \\
D = 2PS^2, & P \equiv 1 \pmod 4, & S, \epsilon = +, -, \\
D = 2PS^2, & P \equiv 3 \pmod 4, & S, \epsilon = -, -;
\end{array}$$

n, n' are any positive numbers prime to $2D$, $\left(\frac{D}{n}\right)$ is Legendre's symbol as generalized by Jacobi, viz. in general if p be a positive or negative prime not a factor of n, then $\left(\frac{n}{p}\right) = +$ or $-$ according as n is or is not a quadratic residue of p (or, what is the same thing, p being positive, $\left(\frac{n}{p}\right) \equiv n^{\frac{1}{2}(p-1)} \pmod p$), and for $P = pp'p'' \ldots$,

$$\left(\tfrac{n}{P}\right) \;=\; \left(\frac{n}{p}\right)\left(\frac{n}{p'}\right)\left(\frac{n}{p''}\right)\cdots$$

and the summation extends to all the values of n, n' of the form above mentioned. In the particular case $D = -1$, it is necessary that the second side should be doubled.

The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$D=-1, \quad \Sigma\, q^{x^2+y^2} \ = \ 4\,\Sigma\big(\tfrac{-1}{n}\big)\ q^n,$

or $\qquad 2\big(1{+}2q{+}2q^4{+}2q^9{+}\cdots\big)\big(q{+}q^4{+}q^9{+}q^{16}{+}\cdots\big) \ = \ \dfrac{q}{1-q}-\dfrac{q^3}{1-q^3}+\dfrac{q^5}{1-q^5}-\dfrac{q^7}{1-q^7}+\&\text{c.}$

$D=-2, \quad \Sigma\, q^{x^2+2y^2} \ = \ 2\,\Sigma\big(\tfrac{-2}{n}\big)\ q^n,$

or $\qquad \big(1{+}2q^2{+}2q^8{+}2q^{18}\cdots\big)\big(q{+}q^3{+}q^9{+}q^{17}\cdots\big) \ = \ \dfrac{q}{1-q^2}+\dfrac{q^3}{1-q^6}-\dfrac{q^5}{1-q^{10}}-\&\text{c.}$

an example given in the memoir.

$D=-3,$

$$(q{+}q^4{+}q^9{+}q^{16}{+}q^{25}\cdots)\big(1{+}2q^2{+}2q^6{+}2q^{12}\cdots\big)\,+\,2\,(q^2{+}q^3{+}q^6{+}q^{12}{+}\cdots)(q^4{+}q^9{+}q^{16}{+}q^{25}\cdots)$$

$$= \ \frac{q}{1-q^2}-\frac{q+q^2}{1-q^3}+\frac{q^3-q^4}{1-q^6}+\frac{q^7-q^8}{1-q^9}+\&\text{c.}$$

I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates; and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

112.

DEMONSTRATION OF A THEOREM RELATING TO THE PRODUCTS OF SUMS OF SQUARES.

[From the Philosophical Magazine, vol. iv. (1852), pp. 515–519.]

Mr Kirkman, in his paper "On Pluquaternions and Homoid Products of Sums of n Squares" (*Phil. Mag.* vol. xxxiii. [1848] pp. 447–459 and 494–509), quotes from a note of mine the following passage: — "The complete test of the possibility of the product of 2^m squares by 2^m squares reducing itself to a sum of 2^m squares is the following — forming the complete systems of triplets for $(2^m - 1)$ things, if eab, ecd, fac, fdb be any four of them, we must have, paying attention to the signs alone,

$$(\pm\, eab)(\pm\, ecd) \;=\; (\pm\, fac)(\pm\, fdb)\,;$$

i.e. if the first two are of the same sign, the last two must be so also, and *vice versa*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things." I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2 + a^2 + b^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + \cdots) \;=\; w_{\prime\prime}^2 + a_{\prime\prime}^2 + b_{\prime\prime}^2 + \cdots\,,$$

where the 2^m quantities $w, a, b, c, \ldots$ and the 2^m quantities $w_1, a_1, b_1, c_1, \ldots$ are given quantities in terms of which the 2^m quantities $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ have to be determined.

Without attaching any meaning whatever to the symbols $a_0, b_0, c_0, \ldots$ I write down the expressions

$$ww_1 + aa_0 + bb_0 + cc_0 + \cdots\,, \qquad w_1 a_0 + a_1 b_0 + b_1 c_0 + \cdots\,,$$

and I multiply as if $a_0, b_0, c_0, \ldots$ really existed, taking care to multiply without making any transposition in the order *inter se* of two symbols a_0, b_0 combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} ww_1 &+ (aw_1 + a_1 w)\, a_0 + (bw_1 + b_1 w)\, b_0 + \cdots \\ &+ aa_1 a_0^2 + bb_1 b_0^2 + \cdots \\ &+ ab_1 a_0 b_0 + a_1 b\, b_0 a_0 + \cdots\,. \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0 b_0 c_0 \ = \ +,$$

means that in the expression of the quasi-product

$$a_0 b_0,\ b_0 c_0,\ c_0 a_0,\ c_0 b_0,\ a_0 c_0,\ b_0 a_0$$

are to be replaced by

$$c_0,\ a_0,\ b_0,\ -a_0,\ -b_0,\ -c_0\,;$$

and that a quasi-equation, such as $a_0 b_0 c_0 = -$, means that in the expression of the quasi-product

$$b_0 c_0,\ c_0 a_0,\ a_0 b_0,\ c_0 b_0,\ a_0 c_0,\ b_0 a_0$$

are to be replaced by

$$-a_0,\ -b_0,\ -c_0,\ a_0,\ b_0,\ c_0\,.$$

It is in the first place clear that the quasi-equation $a_0 b_0 c_0 = +$ may be written in any one of the six forms

$$a_0 b_0 c_0 \ = \ b_0 c_0 a_0 \ = \ c_0 a_0 b_0 \ = \ -a_0 c_0 b_0 \ = \ -c_0 b_0 a_0 \ = \ -b_0 a_0 c_0\,,$$

and so for the quasi-equation $a_0 b_0 c_0 = -$. This being premised, if we form a system of quasi-equations, such as

$$a_0 b_0 c_0, \quad e_0 a_0 b_0, \quad \dots,$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace $a_0^2, b_0^2, \dots$ each by -1, it is clear that the quasi-product will assume the form

$$w_{\prime\prime} w_1 + a_{\prime\prime} a_0 + b_{\prime\prime} b_0 + c_{\prime\prime} c_0 + \cdots\,,$$

$w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \dots$ being determinate functions of $w, a, b, c, \dots$; $w_1, a_1, b_1, c_1, \dots$, homogeneous of the first order in the quantities of each set; the value of $w_{\prime\prime}$ being obviously in every case

$$w_{\prime\prime} \ = \ w w_1 - a a_1 - b b_1 - c c_1 - \cdots\,,$$

and $a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ containing in every case the terms $aw_1 + a_1w$, $bw_1 + b_1w$, $cw_1 + c_1w$, $\ldots$ but the form of the remaining terms depending as well on the triplets entering into the system of quasi-equations as on the values given to the signs $\pm$; the quasi-equations serving, in fact, to prescribe a rule for the formation of certain functions $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$, the properties of which functions may afterwards be investigated.

Suppose, now, that the system of quasi-equations is such that

$$e_0a_0b_0, \quad e_0c_0d_0$$

being any two of its triplets, with a common symbol e_0, there occur also in the system the triplets

$$f_0a_0c_0, \quad f_0d_0b_0, \quad g_0a_0d_0, \quad g_0b_0c_0\,;$$

and suppose that the corresponding portion of the system is

$$e_0a_0b_0 = \epsilon, \quad e_0c_0d_0 = \epsilon', \quad f_0a_0c_0 = \zeta, \quad f_0d_0b_0 = \zeta', \quad g_0a_0d_0 = \iota, \quad g_0b_0c_0 = \iota',$$

where $\epsilon, \zeta, \iota, \epsilon', \zeta', \iota'$ each of them denote one of the signs $+$ or $-$; then $e_{\prime\prime}, f_{\prime\prime}, g_{\prime\prime}$ will contain respectively the terms

$$\begin{aligned}
&\epsilon\,(ab_1 - a_1b) - \epsilon'\,(cd_1 - c_1d), \\
&\zeta\,(ac_1 - a_1c) + \zeta'\,(db_1 - d_1b), \\
&\iota\,(ad_1 - a_1d) + \iota'\,(bc_1 - b_1c)\,;
\end{aligned}$$

and $e_{\prime\prime}^2 + f_{\prime\prime}^2 + g_{\prime\prime}^2$ contains the terms

$$\begin{aligned}
(a^2 + b^2 + c^2 + d^2)&(a_1^2 + b_1^2 + c_1^2 + d_1^2) - a^2a_1^2 - b^2b_1^2 - c^2c_1^2 - d^2d_1^2 \\
+ 2\big[\, &\epsilon\epsilon'\,(ab_1 - a_1b)(cd_1 - c_1d) \\
&+ \zeta\zeta'\,(ac_1 - a_1c)(db_1 - d_1b) \\
&+ \iota\iota'\,(ad_1 - a_1d)(bc_1 - b_1c)\big]\,;
\end{aligned}$$

and by taking account of the terms $ew_1 + e_1w$, $fw_1 + f_1w$, $gw_1 + g_1w$ in $e_{\prime\prime}, f_{\prime\prime}, g_{\prime\prime}$ respectively, we should have had besides in $e_{\prime\prime}^2 + f_{\prime\prime}^2 + g_{\prime\prime}^2$ the terms

$$(e^2 + f^2 + g^2)\,w_1^2 + (e_1^2 + f_1^2 + g_1^2)\,w^2 + 2(ee_1 + ff_1 + gg_1)\,ww_1.$$

Also $w_{\prime\prime}^2$ contains the terms

$$a^2a_1^2 + b^2b_1^2 + c^2c_1^2 + d^2d_1^2 - 2(ee_1 + ff_1 + gg_1)\,ww_1\,;$$

whence it is easy to see that

$$\begin{aligned} & w_{\prime\prime}^2 + a_{\prime\prime}^2 + b_{\prime\prime}^2 + c_{\prime\prime}^2 + \cdots \\ & \quad = (w^2 + a^2 + b^2 + c^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \cdots) \\ & \qquad + 2\Sigma\big[\,\epsilon\epsilon'\,(ab_1 - a_1b)(cd_1 - c_1d) \\ & \qquad\qquad + \zeta\zeta'\,(ac_1 - a_1c)(db_1 - d_1b) \\ & \qquad\qquad + \iota\iota'\,(ad_1 - a_1d)(bc_1 - b_1c)\big], \end{aligned}$$

where the summation extends to all the quadruplets formed each by the combination of two duads such as ab and cd, or ac and db, or ad and bc, i.e. two duads, which, combined with the same common letter (in the instances just mentioned e, or f, or g), enter as triplets into the system of quasi-equations; so that if $\nu = 2^m - 1$, the number of quadruplets is

$$\tfrac{1}{2} \cdot \tfrac{1}{2}(\nu - 1) \cdot \tfrac{1}{2}(\nu - 3) \cdot \nu \cdot 1 \;=\; \tfrac{1}{8}\nu(\nu - 1)(\nu - 3),$$

and the terms under the sign Σ will vanish identically if only

$$\epsilon\epsilon' \;=\; \zeta\zeta' \;=\; \iota\iota'\,;$$

but the relation $\epsilon\epsilon' = \iota\iota'$ is of the same form as the equation $\epsilon\epsilon' = \zeta\zeta'$; hence if all the relations

$$\epsilon\epsilon' \;=\; \zeta\zeta'$$

are satisfied, the terms under the sign Σ vanish, and we have

$$(w_{\prime\prime}^2 + a_{\prime\prime}^2 + b_{\prime\prime}^2 + c_{\prime\prime}^2 + \cdots) \;=\; (w^2 + a^2 + b^2 + c^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \cdots),$$

which is thus shown to be true, upon the suppositions

1. That the system of quasi-equations is such that

   $$e_0a_0b_0, \quad e_0c_0d_0,$$

   being any two of its triplets with a common symbol e_0, there occur also in the system the triplets

   $$f_0a_0c_0, \quad f_0d_0b_0, \quad g_0a_0d_0, \quad g_0b_0c_0\,.$$

2. That for any two pairs of triplets, such as

   $$e\,ab, \;\; e\,cd \quad \text{and} \quad f\,ac, \;\; f\,db\,,$$

   the product of the signs of the triplets of the first pair is equal to the product of the signs of the triplets of the second pair.

In the case of fifteen things a, b, c, ... the triplets may, as appears from Mr Kirkman's paper, be chosen so as to satisfy the first condition; but the second condition involves, as Mr Kirkman has shown, a contradiction; and therefore the product of two sums, each of them of sixteen squares, is not a sum of sixteen squares. It is proper to remark, that this demonstration, although I think rendered clearer by the introduction of the idea of the system of triplets furnishing the rule for the formation of the expressions $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}$, &c., is not in principle different from that contained in Prof. Young's paper "On an Extension of a Theorem of Euler, &c.", *Irish Transactions*, vol. xxi. [1848, pp. 311–341].

113.

NOTE ON THE GEOMETRICAL REPRESENTATION OF THE INTEGRAL $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$.

[From the Philosophical Magazine, vol. v. (1853), pp. 281–284.]

THE equation of a conic passing through the points of intersection of the conics

$$x^2+y^2+z^2 = 0, \qquad ax^2+by^2+cz^2 = 0,$$

is of the form

$$w\,(x^2+y^2+z^2)+ax^2+by^2+cz^2 = 0,$$

where w is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of w a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of w be considered as equal to a constant quantity k, the line is always a tangent to the conic

$$k\,(x^2+y^2+z^2)+ax^2+by^2+cz^2 = 0,$$

and taking $w=p$ for the other value of w, p is a parameter determining the particular tangent, or, what is the same thing, determining the point of contact of this tangent.

The equation of the tangent is easily seen to be

$$x\sqrt{b-c}\,\sqrt{a+k}\,\sqrt{a+p}+y\sqrt{c-a}\,\sqrt{b+k}\,\sqrt{b+p}+z\sqrt{a-b}\,\sqrt{c+k}\,\sqrt{c+p} = 0\,;$$

suppose that the tangent meets the conic $x^2+y^2+z^2=0$ (which is of course the conic corresponding to $w=\infty$) in the points P, P', and let θ, ∞ be the parameters of the point P, and θ', ∞ the parameters of the point P', i.e. (repeating the definition of the terms) let the tangent at P of the conic $x^2+y^2+z^2=0$ be also touched by the conic $\theta\,(x^2+y^2+z^2)+ax^2+by^2+cz^2=0$, and similarly for θ'. The coordinates of the point P are given by the equations

$$x:y:z = \sqrt{b-c}\,\sqrt{a+\theta} \,:\, \sqrt{c-a}\,\sqrt{b+\theta} \,:\, \sqrt{a-b}\,\sqrt{c+\theta}\,;$$

and substituting these values in the equation of the line PP', we have

$$(b-c)\sqrt{a+k}\,\sqrt{a+p}\,\sqrt{a+\theta}+(c-a)\sqrt{b+k}\,\sqrt{b+p}\,\sqrt{b+\theta}+(a-b)\sqrt{c+k}\,\sqrt{c+p}\,\sqrt{c+\theta} = 0, \quad \ldots(*)$$

an equation connecting the quantities p, θ. To rationalize this equation, write

$$\begin{aligned}
\sqrt{(a+k)(a+p)(a+\theta)} &= \lambda + \mu a,\\
\sqrt{(b+k)(b+p)(b+\theta)} &= \lambda + \mu b,\\
\sqrt{(c+k)(c+p)(c+\theta)} &= \lambda + \mu c,
\end{aligned}$$

values which evidently satisfy the equation in question. Squaring these equations, we have equations from which λ^2, $\lambda\mu$, μ^2 may be linearly determined; and making the necessary reductions, we find

$$\begin{aligned}
\lambda^2 &= abc + kp\theta,\\
-2\lambda\mu &= bc + ca + ab - (p\theta + kp + k\theta),\\
\mu^2 &= a + b + c + k + p + \theta,
\end{aligned}$$

or, eliminating λ, μ,

$$\big[bc+ca+ab-(p\theta+kp+k\theta)\big]^2 - 4\,(a+b+c+k+p+\theta)\,(abc+kp\theta) = 0, \quad \dots(*)$$

which is the rational form of the former equation marked $(*)$. It is clear from the symmetry of the formula, that the same equation would have been obtained by the elimination of L, M from the equations

$$\begin{aligned}
\sqrt{(k+a)(k+b)(k+c)} &= L + Mk,\\
\sqrt{(p+a)(p+b)(p+c)} &= L + Mp,\\
\sqrt{(\theta+a)(\theta+b)(\theta+c)} &= L + M\theta,
\end{aligned}$$

and it follows from Abel's theorem (but the result may be verified by means of Euler's fundamental integral in the theory of elliptic functions), that if

$$\Pi x = \int_\infty^x \frac{dx}{\sqrt{(x+a)(x+b)(x+c)}},$$

then the algebraical equations $(*)$ are equivalent to the transcendental equation

$$\pm\Pi k \pm \Pi p \pm \Pi\theta = 0\,;$$

the arbitrary constant which should have formed the second side of the equation having been determined by observing that the algebraical equation gives for $p = \theta$, $k = \infty$, a system of values, which, when the signs are properly chosen, satisfy the transcendental equation. In fact, arranging the rational algebraical equation according to the powers of k, it becomes

$$\begin{aligned} &k^2\,(p-\theta)^2 - 2k\left[p\theta\,(p+\theta) + 2(a+b+c)\,p\theta + (bc+ca+ab)(p+\theta) + 2abc\right] \\ &+ p^2\theta^2 - 2(bc+ca+ab)\,p\theta - 4abc(p+\theta) + b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab \;=\; 0, \qquad (*) \end{aligned}$$

which proves the property in question, and is besides a very convenient form of the algebraical integral. The ambiguous signs in the transcendental integral are not of course arbitrary (indeed it has just been assumed that for $p=\theta$, Πp and $\Pi\theta$ are to be taken with opposite signs), but the discussion of the proper values to be given to the ambiguous signs would be at all events tedious, and must be passed over for the present.

It is proper to remark, that $p=\theta$ gives not only, as above supposed, $k=\infty$, but another value of k, which, however, corresponds to the transcendental equation

$$\pm\Pi k \pm 2\Pi p \;=\; 0\,;$$

the value in question is obviously

$$k \;=\; \frac{p^2 - 2(bc+ca+ab)\,p^2 - 8abc\,p + b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab}{(p+a)(p+b)(p+c)}\,.$$

Consider, in general, a cubic function $ax^3+3bx^2y+3cxy^2+dy^3$, or, as I now write it in the theory of invariants, $(a,b,c,d)\,(x,y)^3$, the Hessian of this function is

$$\big(ac-b^2,\ \tfrac{1}{2}(ad-bc),\ bd-c^2\big)(x,y)^2,$$

and applying this formula to the function $(p+a)(p+b)(p+c)$, it is easy to write the equation last preceding in the form

$$\tfrac{1}{4}\,k \;=\; p-(a+b+c) - \frac{9\ \text{Hessian}\,\{(p+a)(p+b)(p+c)\}}{(p+a)(p+b)(p+c)}\,,$$

which is a formula for the duplication of the transcendent Πx.

Reverting now to the general transcendental equation

$$\pm\Pi k \pm \Pi p \pm \Pi\theta \;=\; 0,$$

we have in like manner

$$\pm\Pi k \pm \Pi p \pm \Pi\theta' \;=\; 0\,;$$

and assuming a proper correspondence of the signs, the elimination of Πp gives

$$\Pi\theta' - \Pi\theta \;=\; \pm 2\,\Pi k,$$

i.e. if the points P, P' upon the conic $x^2 + y^2 + z^2 = 0$ are such that their parameters θ, θ' satisfy this equation, the line PP' will be constantly a tangent to the conic

$$k\,(x^2 + y^2 + z^2) + (ax^2 + by^2 + cz^2) \;=\; 0.$$

Hence also, if the parameters k, k', k'' of the conics

$$\begin{aligned} k\,(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 &= 0, \\ k'(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 &= 0, \\ k''(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 &= 0, \end{aligned}$$

satisfy the equation

$$\Pi k + \Pi k' + \Pi k'' \;=\; 0,$$

there are an infinity of triangles inscribed in the conic $x^2 + y^2 + z^2 = 0$, and the sides of which touch the last-mentioned three conics respectively.

Suppose $2\Pi k = \Pi\kappa$ (an equation the algebraic form of which has already been discussed), then

$$\Pi\theta' - \Pi\theta \;=\; \Pi\kappa,$$

$\theta = \infty$ gives $\theta' = \kappa$; or, observing that $\theta = \infty$ corresponds to a point of intersection of the conics $x^2 + y^2 + z^2 = 0$, $ax^2 + by^2 + cz^2 = 0$, κ is the parameter of the point in which a tangent to the conic $k\,(x^2+y^2+z^2)+ax^2+by^2 + cz^2 = 0$ at any one of its intersections with the conic $x^2 + y^2 + z^2 = 0$ meets the last-mentioned conic. Moreover, the algebraical relation between θ, θ' and k (where, as before remarked, κ is a given function of k) is given by a preceding formula, and is simpler than that between θ, θ' and k.

The preceding investigations were, it is hardly necessary to remark, suggested by a well-known memoir of the late illustrious Jacobi, and contain, I think, the extension which he remarks it would be interesting to make of the principles in such memoir to a system of two conics. I propose reverting to the subject in a memoir to be entitled "Researches on the Porism of the in- and circumscribed triangle." *[This was, I think, never written.]*

114.

ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S EXTENSION OF MALFATTI'S PROBLEM.

[From the Philosophical Transactions of the Royal Society of London, vol. cxlii. for the year 1852, pp. 253–278: Received April 12, — Read May 27, 1852.]

The problem, in a triangle to describe three circles each of them touching the two others and also two sides of the triangle, has been termed after the Italian geometer by whom it was proposed and solved, Malfatti's problem. The problem which I refer to as Steiner's extension of Malfatti's problem is as follows: — "To determine three sections of a surface of the second order, each of them touching the two others, and also two of three given sections of the surface of the second order," a problem proposed in Steiner's memoir, "Einige geometrische Betrachtungen," *Crelle*, t. i. [1826, pp. 161–184]. The geometrical construction of the problem in question is readily deduced from that given in the memoir just mentioned for a somewhat less general problem, viz. that in which the surface of the second order is replaced by a sphere; it is for the sake of the analytical developments to which the problem gives rise, that I propose to resume here the discussion of the problem. The following is an analysis of the present memoir:

§ 1. Contains a lemma which appears to me to constitute the foundation of the analytical theory of the sections of a surface of the second order.

§ 2. Contains a statement of the geometrical construction of Steiner's extension of Malfatti's problem.

§ 3. Is a verification, founded on a particular choice of coordinates, of the construction in question.

§ 4. In this section, referring the surface of the second order to absolutely general coordinates, and after an incidental solution of the problem to determine a section touching three given sections, I obtain the equations for the solution of Steiner's extension of Malfatti's problem.

§ 5. Contains a separate discussion of a system of equations, including as a particular case the equations obtained in the preceding section.

§§ 6 and 7. Contain the application of the formulæ for the general system to the equations in § 4, and the development and completion of the solution.

§ 8. Is an extension of some preceding formulæ to quadratic functions of any number of variables.

§ 1. Lemma relating to the sections of a surface of the second order.

If

$$ax^2 + by^2 + cz^2 + dw^2 + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw = 0$$

be the equation of a surface of the second order, and

$$\mathfrak{A}x^2 + \mathfrak{B}y^2 + \mathfrak{C}z^2 + \mathfrak{D}w^2 + 2\mathfrak{F}yz + 2\mathfrak{G}zx + 2\mathfrak{H}xy + 2\mathfrak{L}xw + 2\mathfrak{M}yw + 2\mathfrak{N}zw = 0$$

the reciprocal equation, the condition that the two sections

$$\lambda x + \mu y + \nu z + \rho w = 0, \qquad \lambda' x + \mu' y + \nu' z + \rho' w = 0,$$

may touch, is

$$\begin{aligned}
&\big(\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + \mathfrak{D}\rho^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu + 2\mathfrak{L}\lambda\rho + 2\mathfrak{M}\mu\rho + 2\mathfrak{N}\nu\rho\big)^{\frac{1}{2}} \\
&\quad \times \big(\mathfrak{A}\lambda'^2 + \mathfrak{B}\mu'^2 + \mathfrak{C}\nu'^2 + \mathfrak{D}\rho'^2 + 2\mathfrak{F}\mu'\nu' + 2\mathfrak{G}\nu'\lambda' + 2\mathfrak{H}\lambda'\mu' + 2\mathfrak{L}\lambda'\rho' + 2\mathfrak{M}\mu'\rho' + 2\mathfrak{N}\nu'\rho'\big)^{\frac{1}{2}} \\
&\quad = \mathfrak{A}\lambda\lambda' + \mathfrak{B}\mu\mu' + \mathfrak{C}\nu\nu' + \mathfrak{D}\rho\rho' + \mathfrak{F}(\mu\nu' + \mu'\nu) + \mathfrak{G}(\nu\lambda' + \nu'\lambda) + \mathfrak{H}(\lambda\mu' + \lambda'\mu) \\
&\qquad + \mathfrak{L}(\lambda\rho' + \lambda'\rho) + \mathfrak{M}(\mu\rho' + \mu'\rho) + \mathfrak{N}(\nu\rho' + \nu'\rho);
\end{aligned}$$

and in particular if the equation of the surface be

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0,$$

the condition of contact is

$$\begin{aligned}
&\big(\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu + K\,\tfrac{\rho^2}{p}\big)^{\frac{1}{2}} \\
&\quad \times \big(\mathfrak{A}\lambda'^2 + \mathfrak{B}\mu'^2 + \mathfrak{C}\nu'^2 + 2\mathfrak{F}\mu'\nu' + 2\mathfrak{G}\nu'\lambda' + 2\mathfrak{H}\lambda'\mu' + K\,\tfrac{\rho'^2}{p}\big)^{\frac{1}{2}} \\
&\quad = \mathfrak{A}\lambda\lambda' + \mathfrak{B}\mu\mu' + \mathfrak{C}\nu\nu' + \mathfrak{F}(\mu\nu' + \mu'\nu) + \mathfrak{G}(\nu\lambda' + \nu'\lambda) + \mathfrak{H}(\lambda\mu' + \lambda'\mu) + K\,\tfrac{\rho\rho'}{p},
\end{aligned}$$

in which last formula

$$\mathfrak{F} = gh - af, \quad \mathfrak{G} = hf - bg, \quad \mathfrak{H} = fg - ch,$$
$$K = abc - af^2 - bg^2 - ch^2 + 2fgh.$$

§ 2.

In order to state in the most simple form the geometrical construction for the solution of Steiner's extension of Malfatti's problem, let the given sections be called for conciseness the *determinators* [1]; any two of these sections lie in two different cones, the vertices of which determine with the line of intersection of the planes of the determinators, two planes which may be termed *bisectors*; the six bisectors pass three and three through four straight lines; and it will be convenient to use the term bisectors to denote, not the entire system, but any three bisectors passing through the same line. Consider three sections, which may be termed *tactors*, each of them touching a determinator and two bisectors, and three other sections (which may be termed *separators*) each of them passing through the point of contact of a determinator and tactor and touching the other two tactors; the separators will intersect in a line which passes through the point of intersection of the determinators. The three required sections, or as I shall term them the *resultors*, are determined by the conditions that each resultor touches two determinators and two separators, the possibility of the construction being implied as a theorem. The *à posteriori* verification may be obtained as follows:—

§ 3.

Let $x = 0$, $y = 0$, $z = 0$ be the equations of the resultors, $w = 0$ the equation of the polar of the point of intersection of the resultors. Since the resultors touch two and two, the equation of the surface is easily seen to be of the form

$$2yz + 2zx + 2xy + w^2 = 0. \qquad (=)$$

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The determinators are sections each of them touching two resultors, but otherwise arbitrary; their equations are

$$-\alpha x + \tfrac{1}{2\alpha}\, y + \tfrac{1}{2\alpha}\, z + w = 0, \qquad \tfrac{1}{2\beta}\, x - \beta y + \tfrac{1}{2\beta}\, z + w = 0, \qquad \tfrac{1}{2\gamma}\, x + \tfrac{1}{2\gamma}\, y - \gamma z + w = 0.$$

[1] I use the words "determinators," &c. to denote indifferently the sections or the planes of the sections; the context is always sufficient to prevent ambiguity.

[2] The reciprocal form is, it should be noted,

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy - 2w^2 = 0.$$

The separators are sections each of them touching two resultors at their point of contact (or what is the same thing, passing through the line of intersection of two resultors), and all of them having a line in common. Their equations may be taken to be

$$cy - bz = 0, \qquad az - cx = 0, \qquad bx - ay = 0.$$

[The memoir continues from § 3 onward at p. 60 et seq., outside the present PDF pp. 51–75 slice.]